

Generation Interconnection System Impact Study Report

For Queue Project Q19-1160

Cottle 138 kV
March 19, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1160. The Interconnection Customer proposes a 148.9 MW solar photovoltaic facility interconnecting at the Cottle 138 kV substation in the Pennsylvania area. The request was received on March 04, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 288 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$115,167,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Rochester 80 to Cttle 138 kV	Wallenpaupack to Hills Chapel 345 kV	147.1	198	37.6
Rochester 80 161 kV/138 kV XFMR	Hills Chael to Wallenpaupack 345 kV L	144.6	174	20.9
Hills Chapel 345-161 kV XFMR	Hills Chapel to Wallenpaupack 345 kV	123.0	311	18.2
Burns 345/138 kV XFMR	Burns to Coronado 345 kV	120.7	478	5.3
Baker Riv to Oak Knoll 138 kV	Burns to Coronado 345 kV L	116.9	484	5.3
Burns to Baker River 138 kV	Burns to Coronado 345 kV	115.5	500	5.3
Hamilton Dome to Oak Knoll Substation 138 kV Line	Burns to Coronado 345 kV	115.0	469	5.3
Rochester 80 Substation to Hills Chapel Substation 161 kV	Wallenpaupack to Hills Chapel 345 kV	114.2	219	20.9
Borden Co to Schumansville 138 kV	Split Rock 345/138kV Transformer	113.4	190	9.6
Split Rock 345 kV/138 kV TX	W Gallatin to Walnut 345 kV	110.7	365	15.5
Verde Mar 2605 to Schumansville 138 kV	Split Rock 345 kV/138 kV Autotransformer	109.0	221	9.6
Flagstaff to Alliant Sbd 9801 Aegon Martha'S Way 138 kV	West Gallatin to Walnut 345 kV line	107.3	354	5.6
Pearl to Alliant Sbd 9801 Aegon Martha'S Way 138 kV	W Gallatin to Walnut 345 kV	102.9	381	5.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rochster 80 to Cottle 138 kV	Rebuild line with higher-capacity conductor	\$22,078,000	100% (\$22,078,000)
Rochester 80 161 kV/138 kV XFMR	Replace transformer with higher-rated unit	\$6,983,000	100% (\$6,983,000)
Hills Chapel 345-161 kV XFMR	Replace transformer with higher-rated unit	\$22,609,000	100% (\$22,609,000)
Burns 345/138kV TX	Replace transformer with higher-rated unit	\$16,312,000	100% (\$16,312,000)
Oak Knoll to Baker Riv 138 kV Line	Rebuild line with higher-capacity conductor	\$4,279,000	100% (\$4,279,000)
Burns to Baker River 138 kV	Rebuild line with higher-capacity conductor	\$4,504,000	100% (\$4,504,000)
Hamilton Dome Substation to Oak Knoll 138 kV L	Rebuild line with higher-capacity conductor	\$2,441,000	100% (\$2,441,000)
Rochester 80 to Hills Chapel 161 kV L	Rebuild line with higher-capacity conductor	\$7,792,000	100% (\$7,792,000)
Borden Co to Schumansville 138 kV L	Rebuild line with higher-capacity conductor	\$3,556,000	100% (\$3,556,000)
Split Rock 345 kV/138 kV XFMR	Replace transformer with higher-rated unit	\$16,591,000	100% (\$16,591,000)
Verde Mar 2605 to Schumansville 138 kV	Reconductor line	\$6,368,000	100% (\$6,368,000)
Alliant Sbd 9801 Aegon Martha'S Way to Flagstaff 138 kV	Reconductor line	\$924,000	100% (\$924,000)
Pearl to Alliant Sbd 9801 Aegon Martha'S Way 138 kV	Reconductor line	\$730,000	100% (\$730,000)
	Total allocated to project		\$115,167,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.